MAXIMUM ENTROPY MODEL

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History

- The concept of Maximum Entropy can be traced back along multiple threads to Biblical times.
- Introduced to NLP area by Berger et. Al. (1996).
- Used in many NLP tasks: MT, Tagging, Parsing, PP attachment, LM, …
- 1981年，托宾的“投资组合理论”使他获得了当年的诺贝尔经济学奖。

- 老头晒太阳

- I see a girl with a telescope

Modeling

The basic idea

- Goal: estimate p

- Choose p with maximum entropy (or "uncertainty") subject to the constraints (or "evidence").

$$H(p) = -\sum_{x \in A \times B} p(x) \log p(x)$$

$$x = (a, b), \quad where \quad a \in A \wedge b \in B$$

Setting

- From training data, collect (a, b) pairs:
  - a: thing to be predicted (e.g., a class in a classification problem)
  - b: the context
  - E.g., POS tagging:
    - a=NN
    - b=the words in a window and previous two tags

- Learn the prob of each (a, b): $p(a, b)$

Features in POS tagging (Ratnaparkhi, 1996)

Condition	Features	
w_i is not rare	$w_i = X$	$\& \ t_i = T$
w_i is rare	X is prefix of w_i, $\lvert X \rvert \leq 4$	$\& \ t_i = T$
	X is suffix of w_i, $\lvert X \rvert \leq 4$	$\& \ t_i = T$
	w_i contains number	$\& \ t_i = T$
	w_i contains uppercase character	$\& \ t_i = T$
	w_i contains hyphen	$\& \ t_i = T$
$\forall \ w_i$	$t_{i-1} = X$	$\& \ t_i = T$
	$t_{i-2}t_{i-1} = XY$	$\& \ t_i = T$
	$w_{i-1} = X$	$\& \ t_i = T$
	$w_{i-2} = X$	$\& \ t_i = T$
	$w_{i+1} = X$	$\& \ t_i = T$
	$w_{i+2} = X$	$\& \ t_i = T$

context (history)

allowable classes

Maximum Entropy

- Why **maximum** entropy?

- Maximize entropy = Minimize commitment

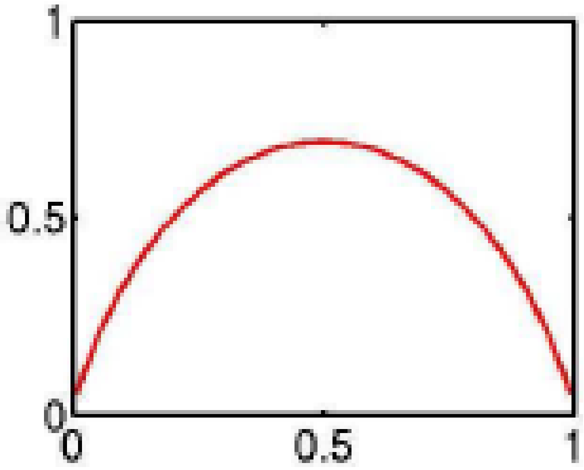
- Model all that is known and assume nothing about what is unknown.
    - Model all that is known: satisfy a set of constraints that must hold
    - Assume nothing about what is unknown:
      choose the most “uniform” distribution
      ➔ choose the one with maximum entropy

COIN-FLIP EXAMPLE

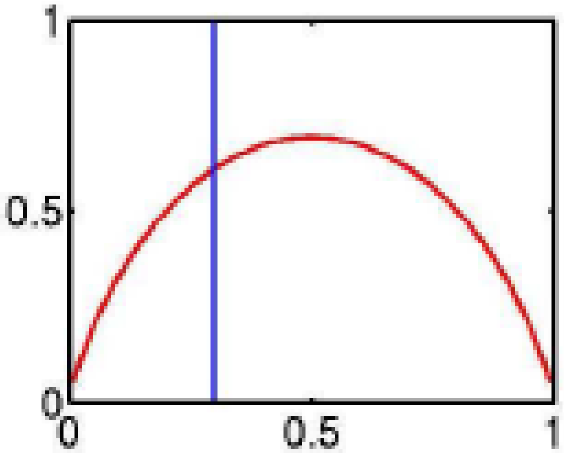
- Toss a coin: p(H)=p1, p(T)=p2.
- Constraint: p1 + p2 = 1
- Question: what's your estimation of p=(p1, p2)?
- Answer: choose the p that maximizes H(p)

$$H(p) = -\sum p(x) \log p(x)$$

H/2

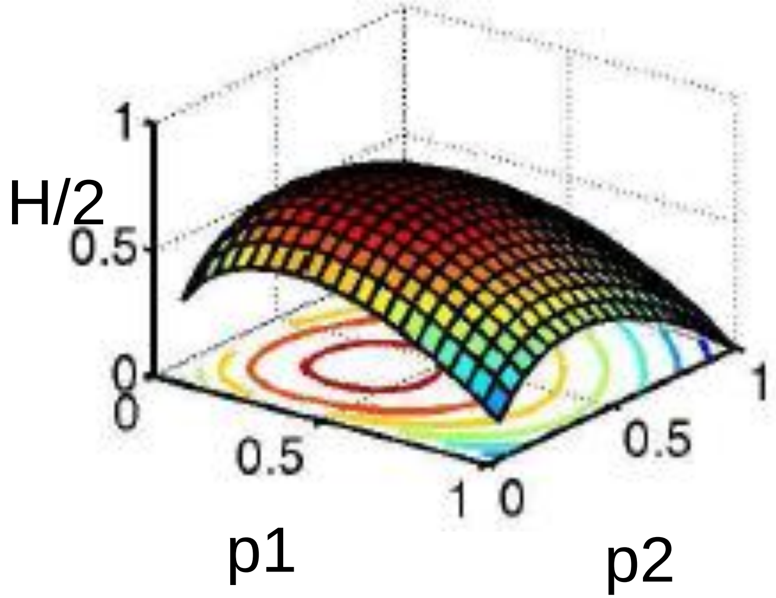


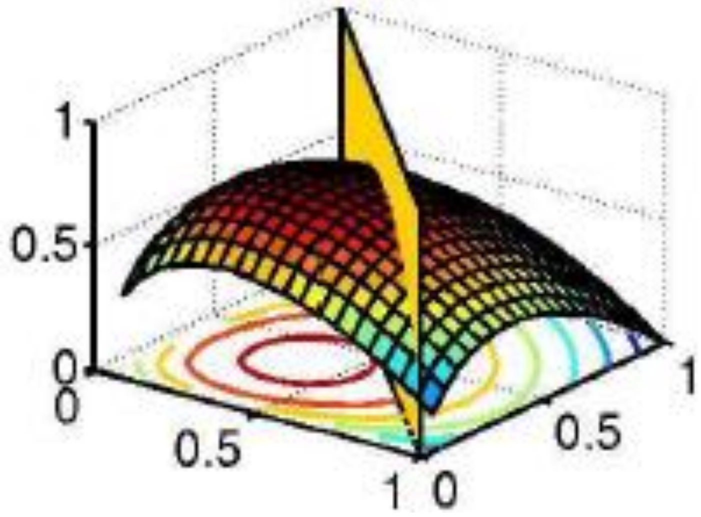
p1



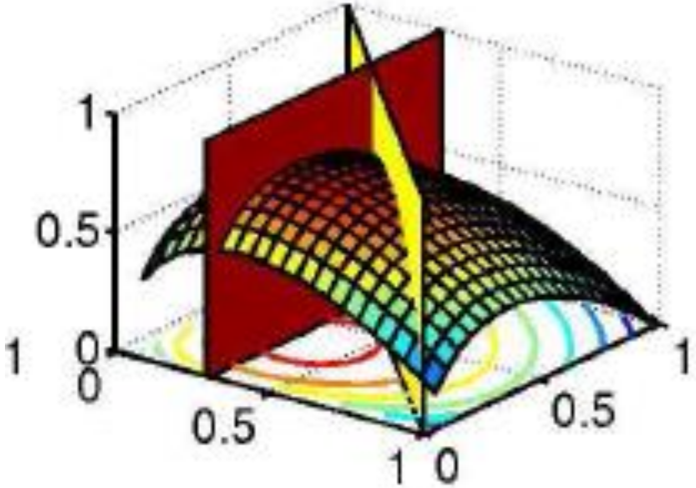
p1=0.3

COIN-FLIP EXAMPLE (cont)





p1 + p2 = 1



p1+p2=1.0, p1=0.3

An MT example(Berger et. al., 1996)

Possible translation for the word "in" is:

$$\{dans, en, \grave{a}, au\ cours\ de, pendant\}$$

Constraint:

$$p(dans) + p(en) + p(\grave{a}) + p(au\ cours\ de) + p(pendant) = 1$$

Intuitive answe

$$\begin{aligned} p(dans) &= 1/5 \\ p(en) &= 1/5 \\ p(\grave{a}) &= 1/5 \\ p(au\ cours\ de) &= 1/5 \\ p(pendant) &= 1/5 \end{aligned}$$

Constraints:

$$
\begin{aligned}
p(dans) + p(en) &= 3/10 \\
p(dans) + p(en) + p(\grave{a}) + p(au\ cours\ de) + p(pendant) &= 1
\end{aligned}
$$

Intuitive answer:

$$
\begin{aligned}
p(dans) &= 3/20 \\
p(en) &= 3/20 \\
p(\grave{a}) &= 7/30 \\
p(au\ cours\ de) &= 7/30 \\
p(pendant) &= 7/30
\end{aligned}
$$

Constraints:

$$\begin{aligned} p(dans) + p(en) &= 3/10 \\ p(dans) + p(en) + p(\grave{a}) + p(au\ cours\ de) + p(pendant) &= 1 \\ p(dans) + p(\grave{a}) &= 1/2 \end{aligned}$$

Intuitive answer:

??

POS tagging example (Klein and Manning, 2003)

Lets say we have the following event space:

NN	NNS	NNP	NNPS	VBZ	VBD

… and the following empirical data:

3	5	11	13	3	1

Maximize H:

$1/e$	$1/e$	$1/e$	$1/e$	$1/e$	$1/e$

… want probabilities: ΣE[NN,NNS,NNP,NNPS,VBZ,VBD] = 1

1/6	1/6	1/6	1/6	1/6	1/6

N* are more common than V*, so we add the feature f_N = {NN, NNS, NNP, NNPS}, with $\tilde{E}[f_N]$ =32/36

NN	NNS	NNP	NNPS	VBZ	VBD
8/36	8/36	8/36	8/36	2/36	2/36

… and proper nouns are more frequent than common nouns, so we add f_P = {NNP, NNPS}, with $\tilde{E}[f_P]$ =24/36

4/36	4/36	12/36	12/36	2/36	2/36

… we could keep refining the models, e.g. by adding a feature to distinguish singular vs. plural nouns, or verb types.

- 例6.1 假设随机变量X有5个取值{A,B,C,D,E}，要估计取各个值的概率P(A), P(B), P(C), P(D), P(E).
  - P(A)+ P(B)+P(C)+P(D)+P(E)=1
  - 从一些先验知识中得到对概率值的约束条件
    P(A)+P(B)=3/10
    P(A)+ P(B)+P(C)+P(D)+P(E)=1
  - 还有第三个约束条件
    P(A)+P(C)=1/2
    P(A)+P(B)=3/10
    P(A)+ P(B)+P(C)+P(D)+P(E)=1

Modeling the problem

- Objective function: H(p)

- Goal: Among all the distributions that satisfy the constraints, choose the one, p*, that maximizes H(p).

$$p^* = \arg\max_{p \in P} H(p)$$

- Question: How to represent constraints?

FEATURES

- Feature (feature function, Indicator function) is a binary-valued function on events:

$$f_j : \varepsilon \rightarrow \{0,1\}, \quad \varepsilon = A \times B$$

- A: the set of possible classes (e.g., tags in POS tagging)
- B: space of contexts (e.g., neighboring words/ tags in POS tagging)
- E.g.:

$$f_j(a,b) = \begin{cases} 1 & if \;\; a = DET \,\&\, curWord(b) = "that" \\ 0 & o.w. \end{cases}$$

SOME NOTATIONS

Finite training sample of events: S

Observed probability of x in S: $\tilde{p}(x)$

The model P's probability of x: $p(x)$

The jth feature: f_j

Observed expectation of f_j :
(empirical count of f_j)

$$E_{\tilde{p}} f_j = \sum_{x \in \varepsilon} \tilde{p}(x) f_j(x)$$

Model expectation of f_j :

$$E_p f_j = \sum_{x \in \varepsilon} p(x) f_j(x)$$

CONSTRAINTS

- Model's feature expectation = observed feature expectation

$$E_p f_j = E_{\tilde{p}} f_j$$

- How to calculate $E_{\tilde{p}} f_j$?

$$E_{\tilde{p}} f_j = \sum_{x \in \varepsilon} \tilde{p}(x) f_j(x) = \frac{\sum_{\iota=1}^{N} f_j(x)}{N}$$

$$f_j(a,b) = \begin{cases} 1 & if \;\; a = DET \;\&\; curWord(b) = "that" \\ 0 & o.w. \end{cases}$$

TRAINING DATA → OBSERVED EVENTS

Word:	the	stories	about	well-heeled	communities	and	developers
Tag:	DT	NNS	IN	JJ	NNS	CC	NNS
Position:	1	2	3	4	5	6	7

$$
\begin{aligned}
w_i &= \texttt{about} && \&\ t_i = \texttt{IN} \\
w_{i-1} &= \texttt{stories} && \&\ t_i = \texttt{IN} \\
w_{i-2} &= \texttt{the} && \&\ t_i = \texttt{IN} \\
w_{i+1} &= \texttt{well-heeled} && \&\ t_i = \texttt{IN} \\
w_{i+2} &= \texttt{communities} && \&\ t_i = \texttt{IN} \\
t_{i-1} &= \texttt{NNS} && \&\ t_i = \texttt{IN} \\
t_{i-2}t_{i-1} &= \texttt{DT NNS} && \&\ t_i = \texttt{IN}
\end{aligned}
$$

RESTATING THE PROBLEM

The task: find *p** s.t. $$p^* = \arg\max_{p \in P} H(p)$$

where $$P = \{ p \mid E_p f_j = E_{\tilde{p}} f_j ,\, j = \{1, \ldots, k\} \}$$

Objective function: *H(p)*

Constraints: $$\{ E_p f_j = E_{\tilde{p}} f_j = d_j ,\, j = \{1, \ldots, k\} \}$$

$$\sum_x p(x) = 1$$

Add a feature $f_0(a,b) = 1 \quad \forall a, b$

$$E_p f_0 = E_{\tilde{p}} f_0 = 1$$

Questions

- What is the definition of p^*?
- What is the form of p^*?
- How to find p^*?

What is the definition of p^*?

- 假设满足所有约束条件的模型集合为$\mathcal{C}$：

$$\mathcal{C} \equiv \{P \in \mathcal{P} | E_P(f_i) = E_{\tilde{P}}(f_i),\ i = 1,2,\cdots,n\}$$

$\mathcal{P}$为没有任何约束条件时所有概率分布的集合

$P(Y|X)$为条件概率分布

集合$\mathcal{P}$中的一个元素，即$P(Y|X) \in \mathcal{P}$

可得出$\mathcal{C}$为$\mathcal{P}$的子集，即f_i为第i个特征函数，一共有n个特征函数。

What is the definition of p*?

定义在条件概率分布$P(Y|X)$上的条件熵为：

$$H(P) = -\sum_{x,y} \tilde{P}(x) P(y|x) \log P(y|x)$$

$\tilde{P}(x)$为边缘分布$P(x)$的经验分布，对数为自然对数。条件熵的最小值为0，最大值为$log|Y|$，其中Y为其可能取值的个数。

则最大熵模型即为在集合$\mathcal{C}$中找出满足以下条件的概率分布：

$$P^* = \underset{P \in \mathcal{C}}{\operatorname{argmax}} H(P)$$

$P^* \in \mathcal{C}$，$H(P)$为一个概率分布的条件熵。

$$p^* = \arg\max_{p \in P} \sum_{(x,y)} p(y \mid x)\tilde{p}(x) \log \frac{1}{p(y \mid x)}$$

$$P = \left\{ p(y \mid x) \;\middle|\; \begin{array}{ll} \forall f_i: & \displaystyle\sum_{(x,y)} p(y \mid x)\tilde{p}(x) f_i(x,y) \\ & = \displaystyle\sum_{(x,y)} \tilde{p}(x,y) f_i(x,y) \\ \forall x: & \displaystyle\sum_{y} p(y \mid x) = 1 \end{array} \right\}$$

定义条件熵 $$H(y|x)=-\sum_{(x,y)\in z} p(y,x)\log p(y|x)$$

模型目标 $$p^{*}(y|x)=\arg\max_{p(y|x)\in P} H(y|x)$$

定义特征函数 $$f_i(x,y)\in\{0,1\}\quad i=1,2,\cdots,m$$

约束条件 $$\sum_{y\in Y} p(y|x)=1$$

（2）

$$E(f_i)=\tilde{E}(f_i)\quad i=1,2,\cdots,m\qquad N=|T|$$

$$\tilde{E}(f_i)=\sum_{(x,y)\in z}\tilde{p}(x,y)f_i(x,y)=\frac{1}{N}\sum_{(x,y)\in T} f_i(x,y)$$

$$E(f_i)=\sum_{(x,y)\in z} p(x,y)f_i(x,y)=\sum_{(x,y)\in z} p(x)p(y|x)f_i(x,y)$$

WHAT IS THE FORM OF P^*? (RATNAPARKHI, 1997)

$$P = \{p \mid E_p f_j = E_{\tilde{p}} f_j ,\ j = \{1,\ldots,k\}\}$$

$$Q = \{p \mid p(x) = \pi \prod_{j=1}^{k} \alpha_j^{f_j(x)}, \alpha_j > 0\}$$

Theorem: if $p^* \in P \cap Q$ then $p^* = \arg\max_{p \in P} H(p)$

Furthermore, p* is unique.

How to find *p**?

- 最大熵模型的学习转化为约束最优化问题：

$$\max_{P \in \mathcal{C}} \quad H(P) = -\sum_{x,y} \tilde{P}(x) P(y|x) \log P(y|x)$$

$$\textbf{s.t.} \ \forall x, y: P(y|x) \geq 0$$

$$\sum_{y} P(y|x) = 1 \text{ for all } x$$

$$E_P(f_i) = E_{\tilde{P}}(f_i), \qquad i = 1, 2, \cdots, n$$

How to find *p**?

- 按照求解最优化问题的习惯，将求最大值问题转化为求解最小值的问题：

$$\min_{p\in\mathcal{C}} \quad -H(P)=\sum_{x,y}\tilde{P}(x)P(y|x)\log P(y|x)$$

$$\textbf{s.t.}\ \ \forall x,y: P(y|x)\geq 0$$

$$\sum_y P(y|x)=1 \text{ for all } x$$

$$E_P(f_i)=E_{\tilde{P}}(f_i), \qquad i=1,2,\cdots,n$$

拉格朗日函数$L(p,\lambda)$

$$
\begin{aligned}
&L(P,\lambda) \\
&= -H(P) + \lambda_0\left(1 - \sum_y P(y|x)\right) \\
&+ \sum_{i=1}^{n} \lambda_i (E_{\tilde{P}}(f_i) - E_P(f_i))
\end{aligned}
$$

$$
= \sum_{x,y} \tilde{P}(x)P(y|x)logP(y/x) + \lambda_0\left(1 - \sum_y P(y|x)\right)
$$

$$
+ \sum_{i=1}^{n} \lambda_i \left(\sum_{x,y} \tilde{P}(x,y) f_i(x,y) - \sum_{x,y} \tilde{P}(x)P(y|x) f_i(x,y)\right)
$$

$$\frac{\partial L(P,\lambda)}{\partial P(y|x)}$$

$$= \sum_{x,y} \tilde{P}(x)(\log P(y|x) + 1) - \sum_{y} \lambda_0$$

$$- \sum_{x,y} \left(\tilde{P}(x) \sum_{i=1}^{n} \lambda_i f_i(x,y) \right)$$

$$= \sum_{x,y} \tilde{P}(x)(\log P(y|x) + 1 - \lambda_0 - \sum_{i=1}^{n} \lambda_i f_i(x,y))$$

令上述偏导数等于0，在$\tilde{\mathrm{P}}(x) > 0$的情况下

$$P(y|x) = \exp(\sum_{i=1}^{n} \lambda_i f_i(x,y) + \lambda_0 - 1) = \frac{\exp(\sum_{i=1}^{n} \lambda_i f_i(x,y))}{\exp(1-\lambda_0)}$$

任意*x*有$\sum_y p(y|x) = 1$，令$1/\exp\,(1-\lambda_0)$为常数*c*

$$\sum_y c e^{\sum_i \lambda_i f_i(x,y)} = 1$$

解得：

$$\mathrm{c} = \frac{1}{\sum_y e^{\sum_i \lambda_i f_i(x,y)}}$$

令

$$Z(x) = \sum_y e^{\sum_i \lambda_i f_i(x,y)}$$

$$P_{\lambda}(y|x) = \frac{1}{Z_{\lambda}(x)} \exp(\textstyle\sum_{i=1}^{n} \lambda_i f_i(x, y))$$

$$Z_{\lambda}(x) = \sum_{y} \exp\left(\sum_{i=1}^{n} \lambda_i f_i(x, y)\right)$$

- 例6.2 假设随机变量X有5个取值{A,B,C,D,E}，要估计取各个值的概率P(A), P(B), P(C), P(D), P(E).从一些先验知识中得到对概率值的约束条件 $P(A)+P(B)=3/10$。

例 6.2 学习例 6.1 中的最大熵模型.

解 为了方便，分别以 y_1, y_2, y_3, y_4, y_5 表示 A，B，C，D 和 E，于是最大熵模型学习的最优化问题是

$$\min \ -H(P) = \sum_{i=1}^{5} P(y_i) \log P(y_i)$$

$$\text{s.t.} \quad P(y_1) + P(y_2) = \tilde{P}(y_1) + \tilde{P}(y_2) = \frac{3}{10}$$

$$\sum_{i=1}^{5} P(y_i) = \sum_{i=1}^{5} \tilde{P}(y_i) = 1$$

引进拉格朗日乘子 w_0, w_1，定义拉格朗日函数

$$L(P, w) = \sum_{i=1}^{5} P(y_i) \log P(y_i) + w_1 \left(P(y_1) + P(y_2) - \frac{3}{10} \right) + w_0 \left(\sum_{i=1}^{5} P(y_i) - 1 \right)$$

根据拉格朗日对偶性，可以通过求解对偶最优化问题得到原始最优化问题的解，所以求解

$$\max_{w}\min_{P} L(P,w)$$

首先求解 $L(P,w)$ 关于 P 的极小化问题．为此，固定 w_0, w_1，求偏导数：

$$\frac{\partial L(P,w)}{\partial P(y_1)} = 1+\log P(y_1)+w_1+w_0$$

$$\frac{\partial L(P,w)}{\partial P(y_2)} = 1+\log P(y_2)+w_1+w_0$$

$$\frac{\partial L(P,w)}{\partial P(y_3)} = 1+\log P(y_3)+w_0$$

$$\frac{\partial L(P,w)}{\partial P(y_4)} = 1+\log P(y_4)+w_0$$

$$\frac{\partial L(P,w)}{\partial P(y_5)} = 1+\log P(y_5)+w_0$$

令各偏导数等于 0，解得

$$P(y_1)=P(y_2)=\mathrm{e}^{-w_1-w_0-1}$$

$$P(y_3)=P(y_4)=P(y_5)=\mathrm{e}^{-w_0-1}$$

于是，

$$\min_{P} L(P,w)=L(P_w,w)=-2\mathrm{e}^{-w_1-w_0-1}-3\mathrm{e}^{-w_0-1}-\frac{3}{10}w_1-w_0$$

于是，

$$\min_P L(P,w)=L(P_w,w)=-2\mathrm{e}^{-w_1-w_0-1}-3\mathrm{e}^{-w_0-1}-\frac{3}{10}w_1-w_0$$

再求解 $L(P_w,w)$ 关于 w 的极大化问题：

$$\max_w\ L(P_w,w)=-2\mathrm{e}^{-w_1-w_0-1}-3\mathrm{e}^{-w_0-1}-\frac{3}{10}w_1-w_0$$

分别求 $L(P_w,w)$ 对 w_0, w_1 的偏导数并令其为 0，得到

$$\mathrm{e}^{-w_1-w_0-1}=\frac{3}{20}$$

$$\mathrm{e}^{-w_0-1}=\frac{7}{30}$$

于是得到所要求的概率分布为

$$P(y_1)=P(y_2)=\frac{3}{20}$$

$$P(y_3)=P(y_4)=P(y_5)=\frac{7}{30}$$

对偶函数的极大化等价于最大熵的极大似然估计

似然函数与极大似然估计

- 设随机样本$X_1, X_2, \ldots, X_n$来自分布密度（或概率函数）为$f(X, \theta)$的总体，其中θ是未知参数，则

$$L(\theta; X_1, \ldots, X_n) = f(X_1, \theta) \ldots f(X_n, \theta)$$

称为样本$(X_1, X_2, \ldots, X_n)$的似然函数（likelihood function），简记为$L(\theta)$。

似然函数与极大似然估计

- 使似然函数$L(\theta)$取最大值的$\hat{\theta}$，即

$$L(\hat{\theta}) = \operatorname*{Sup}_{\theta} L(\theta)$$

称为θ的极大似然估计（maximum likelihood estimator，MLE），记之为$\hat{\theta}_{\mathrm{MLE}}$，等价于求$\hat{\theta}$。

- 极大似然估计是在给定模型（含有未知参数）和数据集的情况下，估计出最佳的模型中未知参数的值，这些参数的取值可以使模型实现对给定数据集最大程度的拟合。

似然函数与极大似然估计

- 使得

$$\ln L(\hat{\theta}) = \max_{\theta} \ln L(\theta)$$

可得

$$\frac{\mathrm{d}\ln L(\theta)}{\mathrm{d}\theta} = 0$$

称为似然方程。

- 当有多个参数时，上述似然方程变为方程组

$$\frac{\partial \ln L(\theta_1, \ldots, \theta_k)}{\partial \theta_i} = 0, i \leq k$$

其中k为参数的个数。

- 例：给定一个硬币，此硬币有正面和反面两个不同的面，使用此硬币进行10次的抛硬币实验，最终得到的10次实验的结果为：正面，正面，反面，正面，正面，正面，反面，反面，正面，正面。令每次抛硬币结果为正面的概率为 p ，则根据极大似然估计方法，算出p的取值。

解：

由题意可知，每次抛硬币结果为反面的概率为1-p。

由于每次抛硬币实验是相互独立的，所以得到上述实验结果的概率为：

$$\mathrm{P} = pp(1-p)ppp(1-p)(1-p)pp = p^7(1-p)^3$$

极大似然估计就是要找出与样本的分布最接近的概率分布模型，因此要求上式取得最大值，即：

$$\max P = \max_{0 \le p \le 1} p^7(1-p)^3$$

对参数***p***进行求导，并令其求导之后的算式为0，即可求出上式最优解为***p***=0.7。

- 例：考虑投掷一枚硬币的实验。假如已知投出的硬币正面朝上的概率是$P_H = 0.5$，便可知投掷若干次后出现各种结果的可能性。

投两次都是正面朝上的概率是0.25

$$P_{HH} = 0.25$$

即给定“投两次都是正面朝上”的观测，则硬币正面朝上的概率为0.5的似然是

$$L(p_H = 0.5; HH) = P(HH|p_H = 0.5) = 0.25$$

如果考虑 P_H= 0.6，那么似然函数的值会变大

$$L(p_H = 0.6; HH) = 0.36.$$

如果参数的取值变成0.6的话，结果观测到连续两次正面朝上的概率要比假设0.5 时更大。

参数取成0.6 要比取成0.5 更有说服力，更为“合理”。

- 似然函数的重要性不是它的具体取值，而是当参数变化时函数到底变小还是变大。
- 对同一个似然函数，如果存在一个参数值，使得它的函数值达到最大的话，那么这个值就是最为“合理”的参数值。

最大熵模型中条件概率分布$P(y|x)$对数似然函数为：

$$L_{\tilde{P}}(P) = \log \prod_{x,y} P(y|x)^{\tilde{P}(x,y)} = \sum_{x,y} \tilde{P}(x,y) \log P(y|x)$$

其中$\tilde{P}(x,y)$为已知的从训练数据中得到的经验概率分布。

$$P_{\wedge}(y|x) = \frac{1}{Z_{\wedge}(x)} \exp\left(\sum_{i=1}^{n} \wedge_i f_i(x,y)\right)$$

$$Z_{\wedge}(x) = \sum_{y} \exp\left(\sum_{i=1}^{n} \wedge_i f_i(x,y)\right)$$

$$L_{\tilde{P}}\left(P_{\wedge}\right) = \sum_{x,y} \tilde{P}(x,y) \log P(y|x)$$

$$= \sum_{x,y} \tilde{P}(x,y) \left(\sum_{i=1}^{n} \wedge_i f_i(x,y) - \log Z_{\wedge}(x)\right)$$

$$= \sum_{x,y} \tilde{P}(x,y) \sum_{i=1}^{n} \lambda_i f_i(x,y) - \sum_{x,y} \tilde{P}(x,y) \log Z_{\lambda}(x)$$

$$= \sum_{x,y} \tilde{P}(x,y) \sum_{i=1}^{n} \lambda_i f_i(x,y) - \sum_{x} \tilde{P}(x) \left(\sum_{y} P_{\lambda}(y|x) \right) \log Z_{\lambda}(x)$$

$$= \sum_{x,y} \tilde{P}(x,y) \sum_{i=1}^{n} \lambda_i f_i(x,y) - \sum_{x} \tilde{P}(x) \log Z_{\lambda}(x)$$

$Z_{\lambda}(x) = \sum_y e^{\sum_i \lambda_i f_i(x,y)}$，并且$\tilde{P}(x,y) = \tilde{P}(x)P(y|x)$

$$\max_{P\in \mathbf{C}} \quad H(P) = -\sum_{x,y} \tilde{P}(x)P(y|x)\log P(y|x)$$

$$\text{s.t.} \quad E_P(f_i) = E_{\tilde{P}}(f_i), \quad i = 1,2,\cdots,n$$

$$\sum_y P(y|x) = 1$$

$$\min_{P\in \mathbf{C}} \quad -H(P) = \sum_{x,y} \tilde{P}(x)P(y|x)\log P(y|x)$$

$$\text{s.t.} \quad E_P(f_i) - E_{\tilde{P}}(f_i) = 0\,, \quad i = 1,2,\cdots,n$$

$$\sum_y P(y|x) = 1$$

$$L(P,w) \equiv -H(P) + w_0\left(1 - \sum_y P(y|x)\right) + \sum_{i=1}^{n} w_i (E_{\tilde{P}}(f_i) - E_P(f_i))$$

$$= \sum_{x,y} \tilde{P}(x)P(y|x)\log P(y|x) + w_0\left(1 - \sum_y P(y|x)\right)$$

$$+ \sum_{i=1}^{n} w_i \left(\sum_{x,y} \tilde{P}(x,y) f_i(x,y) - \sum_{x,y} \tilde{P}(x)P(y|x) f_i(x,y)\right)$$

$$\min_{P\in \mathbf{C}} \max_w L(P,w)$$

$$\max_w \min_{P\in \mathbf{C}} L(P,w)$$

$$\Psi(w) = \min_{P\in \mathbf{C}} L(P,w) = L(P_w, w)$$

$\Psi(w)$ 称为对偶函数．同时，将其解记作

$$P_w = \arg\min_{P\in \mathbf{C}} L(P,w) = P_w(y|x)$$

对于对偶函数$\Psi(w)$

$$\Psi(\wedge)=\sum_{x,y}\tilde{P}(x)P_{\wedge}(y|x)\log P_{\wedge}(y|x)$$

$$+\sum_{i=1}^{n}\wedge_i\left(\sum_{x,y}\tilde{P}(x,y)f_i(x,y)-\sum_{x,y}\tilde{P}(x)P_{\wedge}(y|x)f_i(x,y)\right)$$

$$=\sum_{x,y}\tilde{P}(x,y)*\sum_{i=1}^{n}\wedge_i f_i(x,y)+\sum_{x,y}\tilde{P}(x)P_{\wedge}(y|x)\left(\log P_{\wedge}(y|x)-\sum_{i=1}^{n}\wedge_i f_i(x,y)\right)$$

$$=\sum_{x,y}\tilde{P}(x,y)\sum_{i=1}^{n}\wedge_i f_i(x,y)-\sum_{x,y}\tilde{P}(x)P_{\wedge}(y|x)\log Z_{\wedge}(x)$$

$$=\sum_{x,y}\tilde{P}(x,y)\sum_{i=1}^{n}\wedge_i f_i(x,y)-\sum_{x}\tilde{P}(x)\log Z_{\wedge}(x)$$

其中最后一步用到了$\sum_y P(y|x)=1$。

可以得到：

$$\Psi(\wedge)=L_{\tilde{P}}(P_{\wedge})$$

Summary

- 最大熵模型学习中的对偶问题极大化等价于极大似然估计。
- 最大熵的解是符合数据集中数据分布的最优解，证明了最大熵模型的合理性。

THANK YOU!